CÁSSIO CIONI CARMO

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RESUMO

Electrical engineer with experience in embedded software development since 2020. I currently develop and maintain C software for automotive ECUs. I also have previous experience developing firmware for autonomous agricultural spraying drones, as well as with bare-metal systems, hardware/software integration, low-level debugging, and system validation. I work with protocols such as CAN, SPI, I2C, and UART, focusing on reliability and fault diagnosis.

EDUCATION

Postgraduate Specialization in Embedded Systems for the Automotive Industry

UTFPR, Curitiba, PR — Mar. 2024 - Mar. 2026

Postgraduate Specialization in Industrial Automation

UTFPR, Curitiba, PR — Mar. 2024 - Mar. 2026

BSE, Electrical Engineering

UFSCar, São Carlos, SP — Mar. 2013 -Sept. 2023

TECHNICAL SKILLS

Languages: Advanced English
Programming Languages: C, C++, Python
Embedded Systems: Bare-metal, hardware/software integration, low-level debugging, real-time systems
Tools and Environments: Git, IBM Rational Synergy, VS Code, MATLAB, LabVIEW, LaTeX
Microcontrollers: ESP8266, ESP32, STM32, ATmega
Protocols: CAN, SPI, I2C, UART, WiFi, Bluetooth, ZigBee

PROFESSIONAL EXPERIENCE

Marelli, Hortolândia, SP: Product Analyst — Jul 2024 - Present

- Develop and maintain embedded C software for automotive ECUs used in combustion engines.
- Work on projects for automakers such as Stellantis and Caoa Chery.
- Analyze and fix issues in legacy software, focusing on reliability, functional safety, and performance.
- Integrate and validate software with automotive hardware, including sensors, actuators, and control modules.
- Support testing, debugging, and validation activities in an industrial environment.

Arys, São Carlos, SP: Embedded Software Developer — Oct 2022 - Apr. 2024

- Developed firmware in C/C++ for autonomous agricultural spraying drones.
- Integrated embedded systems involving sensors, actuators, flight controllers, and communication systems.
- Implemented and maintained communication protocols (CAN, SPI, I2C, UART).
- Performed low-level debugging and validation of new hardware.
- Analyzed flight data to assess performance and support continuous system improvement.
- Supported field testing and resolution of critical issues during operation.

Scenario Automation, São Carlos, SP: System Analyst (Embedded Software) — Mar 2021 – Sept 2022

- Developed embedded software for home automation systems.
- Implemented firmware for electric actuation, data acquisition, and wireless communication.
- Integrated microcontrollers with sensors and actuators for IoT applications.
- Maintained and enhanced existing software, focusing on stability and security.
- Used C, C++, and Python in embedded environments.

XMobots, São Carlos, SP: Intern R&D (Embedded Software) — Jan. 2020 – Jan. 2021

- Developed embedded software for drones and ground stations.

- Implemented data acquisition and transmission systems using WiFi, Bluetooth, and LoRa.
- Integrated GNSS modules and post-processed data using PPP, PPK, and RTK techniques.
- Tested and validated new hardware, supporting R&D activities.
- Programmed in C, Python, MicroPython, and MATLAB for embedded systems.

ACADEMIC PROJECTS

Detection and identification of leaf diseases using neural networks

Electrical Engineering capstone project

- Developed software to identify and classify diseases in soybean leaves using image processing.

BAJA UFSCar

Extension project focused on building an off-road car.

- Designed and manufactured PCBs.
- Developed telemetry software.

ZigBee-based Radiolocation

Undergraduate research project.

- Located a ZigBee transmitter based on signal strength received by three other transmitters at known positions.